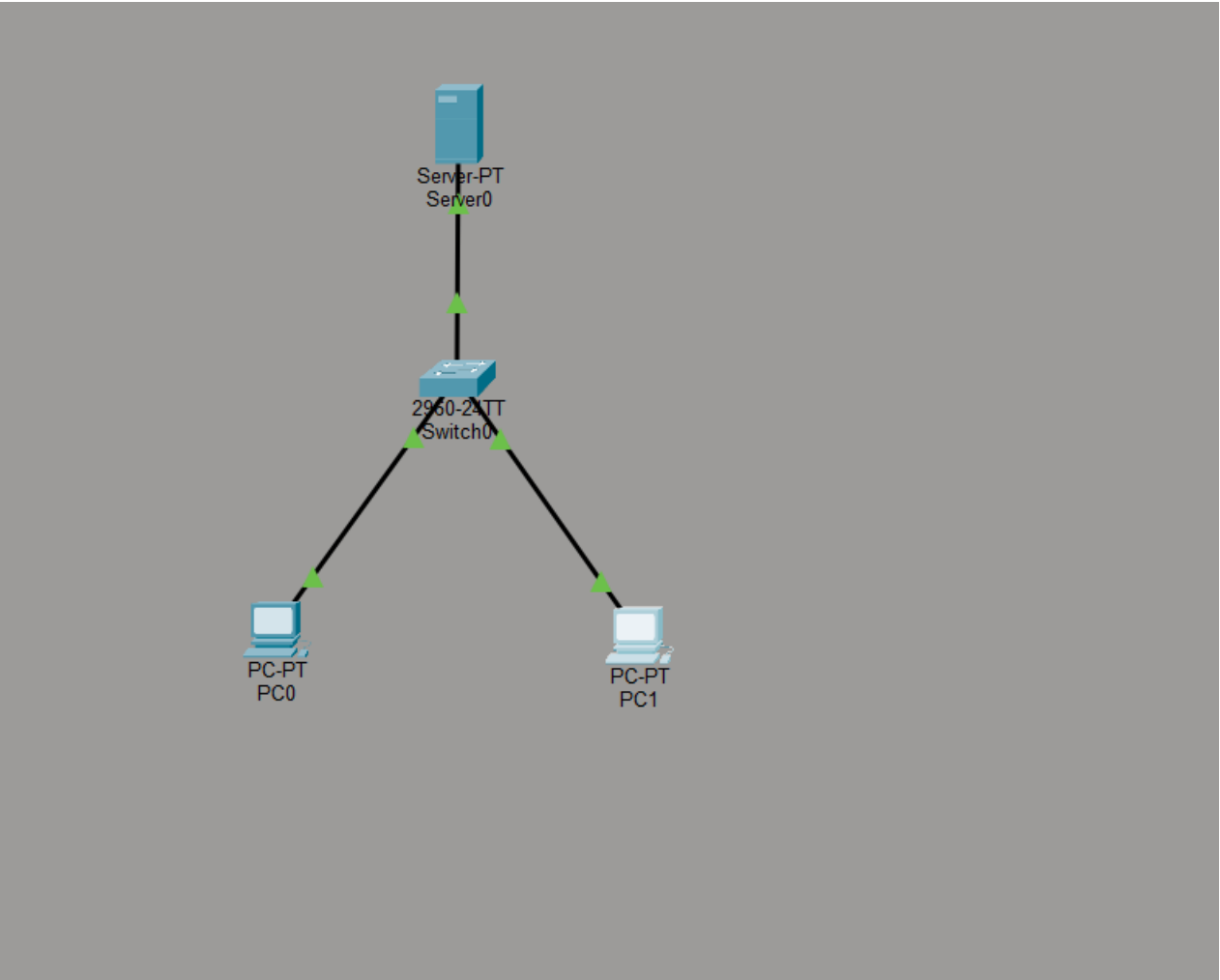


Q-In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

To configure a DHCP server in Cisco Packet Tracer and verify that two client PCs automatically receive different IP addresses from the DHCP pool.

Configuration Steps

1. Placed one server, one switch, and two PCs in the workspace.
2. Connected all devices to the switch using Copper Straight-Through cables.
3. Configured **Server0** with a static IP address:
 - IP Address: **192.168.1.2**
 - Subnet Mask: **255.255.255.0**
 - Default Gateway: **192.168.1.1**
4. Enabled the DHCP service on the server.
5. Created a DHCP pool with the following settings:
 - Pool Name: **LAN**
 - Default Gateway: **192.168.1.1**
 - DNS Server: **8.8.8.8**
 - Start IP Address: **192.168.1.100**
 - Subnet Mask: **255.255.255.0**
 - Maximum Users: **50**
6. Configured both PCs to obtain IP addresses automatically using DHCP.
7. Verified the assigned IP addresses using the **ipconfig** command.
8. Tested network connectivity by sending a ping from **PC0** to **PC1**.



SERVER DHCP:

Physical Config Services Desktop Programming Attributes

SERVICES

DHCP

Interface: FastEthernet0 Service: On

Pool Name: serverPool

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

Start IP Address: 192.168.1.168 Subnet Mask: 255.255.255.0

Maximum Number of Users: 512

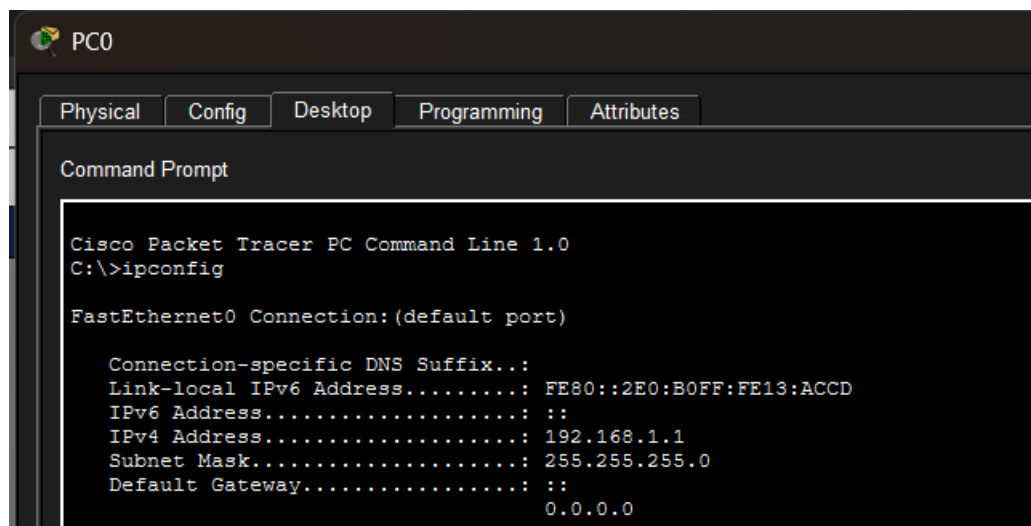
TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Buttons: Add, Save, Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
LAN	192.168.1.1	8.8.8.8	192.168.1.100	255.255.255.0	50	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.1.0	255.255.255.0	512	0.0.0.0	0.0.0.0

PC0 DHCP IP:



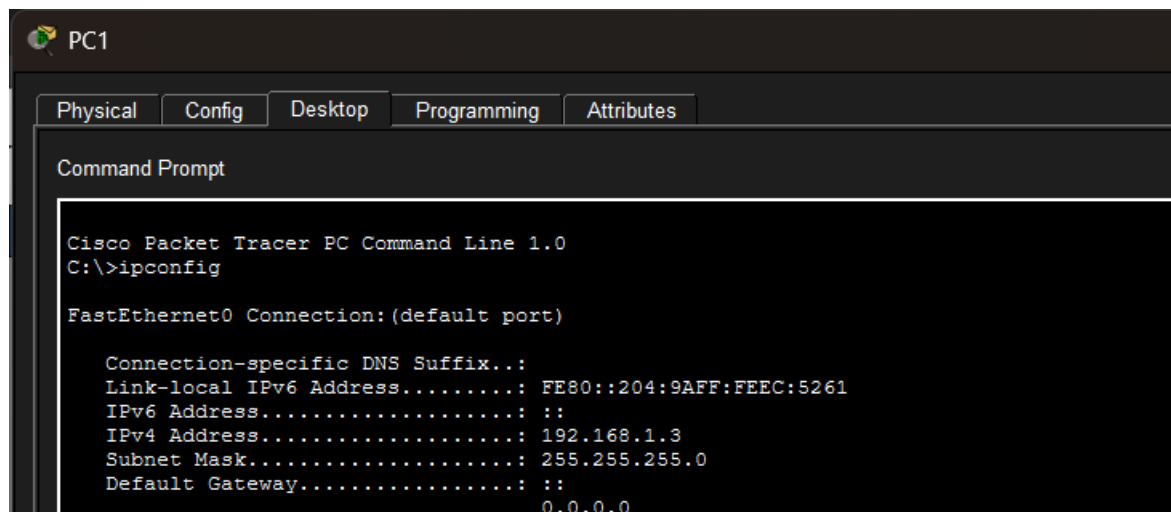
The screenshot shows the 'PC0' window in Cisco Packet Tracer. The 'Config' tab is selected, and the 'Command Prompt' is open. The output of the 'ipconfig' command is displayed, showing the configuration for the 'FastEthernet0' interface.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::2E0:B0FF:FE13:ACCD
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.1
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                                0.0.0.0
```

PC1 DHCP IP:



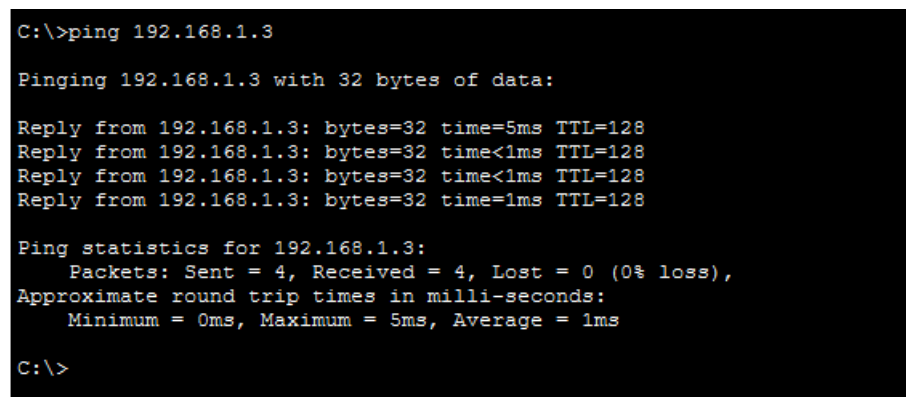
The screenshot shows the 'PC1' window in Cisco Packet Tracer. The 'Config' tab is selected, and the 'Command Prompt' is open. The output of the 'ipconfig' command is displayed, showing the configuration for the 'FastEthernet0' interface.

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::204:9AFF:FEEC:5261
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.3
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                                0.0.0.0
```

Testing:



The screenshot shows a Command Prompt window with the output of a ping command to 192.168.1.3. The output shows four successful replies with a 0% loss rate.

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=5ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>
```